

RAHUL NARAYANAN

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PROFESSIONAL SUMMARY

Final-year Bachelor of Engineering (Honours) student in Mechatronic Engineering at the University of Technology Sydney, graduating in November 2026 and on track for First Class Honours. Experience in robotics, embedded software and autonomous systems through industry and project work using ESP32, Raspberry Pi, ROS 2, computer vision and sensor integration. At Optik Consultancy, helped transform a non-portable AU\$70,000 respiratory-testing system into an accurate handheld prototype costing under AU\$300. Seeking early-career robotics and autonomous systems roles focused on embedded software and systems integration.

EDUCATION

Bachelor of Engineering (Honours) in Mechatronic Engineering
University of Technology Sydney

(WAM: 75.67)
(Feb 2024 – Nov 2026)

EXPERIENCE

Embedded Software Assistant

PAVEMENT MANAGEMENT SERVICES

Aug 2026 – Present

- Defining software and functional requirements for the Phase 1 modular rooftop camera system, including camera input, data handling and compute-platform requirements.
- Researching and evaluating NVIDIA Jetson edge-computing platforms and camera hardware based on processing requirements, resolution, frame rate, connectivity and software compatibility.
- Supporting development of the embedded software architecture and hardware-software interfaces between cameras, NVIDIA Jetson edge-compute hardware and supporting modules.

Engineering Intern | Embedded Software Lead

OPTIK CONSULTANCY | Client: ROYAL NORTH SHORE HOSPITAL

May 2026 – Aug 2026

- Led development of the ESP32 embedded software architecture for a portable respiratory-testing prototype, structuring firmware into modular components for touchscreen control, sensor acquisition and processing, test sequencing, system control and data storage.
- Designed and implemented finite state machine-based interface logic governing touchscreen inputs, screen transitions and complete respiratory-test workflows throughout the prototype operating sequence.
- Integrated sensor acquisition, touchscreen control and data-logging functionality within the ESP32 firmware and supported testing of the complete respiratory-test workflow.

Achievement: Helped transform a non-portable AU\$70,000 respiratory-testing system into an accurate, portable handheld prototype with a build cost of under AU\$300.

Christmas Casual

MYER

Oct 2023 – Feb 2024

- Delivered customer service and processed transactions accurately during high-volume Christmas trading, balancing competing priorities while maintaining clear communication, reliability and attention to detail.

Warehouse Assistant

INDUSTRIAL ENGRAVING SOLUTIONS

Sep 2023

- Prepared, inspected and packaged high-volume solar-panel labels to tight production deadlines, identifying defects and maintaining quality, accuracy and workplace safety standards.

PROJECTS

Follow-Me Cooling Cart | Individual Capstone Project

Feb 2026 – Present

Developing an autonomous battery-powered cooling cart for reliable user tracking and stable transport in outdoor environments.

- Designing a multi-sensor user-tracking and localisation system using LiDAR and computer vision to handle dynamic outdoor conditions.
- Developing real-time path planning, obstacle avoidance and motion control on Raspberry Pi for consistent system performance.

Tech Stack: ROS 2, Raspberry Pi, LiDAR, YOLO, Python, PWM Motor Control, Ubuntu 22.04

Osprey Chess Engine | Personal Project

Jun 2026 – Present

Self-initiated C++ chess engine project focused on intelligent decision-making, search optimisation, performance evaluation and iterative engine development.

- Developing Osprey, a C++ UCI chess engine integrated with a Python application, implementing iterative deepening, Principal Variation Search, alpha-beta pruning, quiescence search, transposition tables and advanced move-ordering heuristics.
- Benchmarking and optimising engine performance through automated testing, perf validation and self-play, achieving wins against a ~1900 Elo chess bot while continuing to improve playing strength against Stockfish.

Tech Stack: C++, Python, Multithreading, AVX2, Bitboards, NNUE, UCI, NumPy, python-chess, Tkinter

Wearable Fitness Tracker | Team Project | High Distinction

Feb 2026 – Jun 2026

Developed a wearable fitness tracker integrating custom PCB hardware, embedded firmware, sensor-based step tracking and real-time user interface features.

- Designed embedded software architecture using a finite state machine to manage step tracking, calibration, self-test, stopwatch, health, game and settings modes.
- Implemented ADXL335-based step detection using calibrated acceleration data, magnitude calculation, hysteresis thresholding and cooldown filtering to reduce false counts.

Tech Stack: C++, Arduino Framework, ESP32, ADXL335, OLED Display, Altium Designer, PCB Design, GitHub

Follow-Me TurtleBot | Team Project | High Distinction

Oct 2025 – Nov 2025

Developed an autonomous TurtleBot for human following using vision and LiDAR-based obstacle avoidance.

- Developed a human-following system using YOLOv8 computer vision and LiDAR sensing for real-time target detection and tracking.
- Implemented obstacle avoidance and tracking logic to maintain stable navigation while following a moving user.

Tech Stack: ROS 1, YOLOv8, LiDAR, Ubuntu 20.04 on WSL, MATLAB

Car Safety System | Individual Project | High Distinction

Oct 2024 – Nov 2024

Built an embedded prototype for real-time obstacle detection and driver attention alert.

- Designed and developed an Arduino-based system integrating ultrasonic sensing, distance measurement and real-time alert generation.
- Implemented visual and audible warning logic using LCD, LED and buzzer outputs to improve driver awareness.

Tech Stack: Arduino, Ultrasonic Sensors, Potentiometer, LCD, LEDs, Buzzers

SKILLS

SOFTWARE – Python, C++, MATLAB/Simulink

ROBOTICS & AUTONOMOUS SYSTEMS – ROS 1, ROS 2, Nav2, Sensor Fusion, Path Planning

EMBEDDED & ELECTRONICS – Arduino, STM32, Raspberry Pi, ESP32, NVIDIA Jetson, PWM Motor Control, PCB Design and Assembly, Electronic Testing

TOOLS – Git, Ubuntu Linux, Altium Designer, WSL

AI & COMPUTER VISION – YOLO, OpenCV, DeepSORT

SYSTEMS & VALIDATION – Requirements Definition, System Integration, Verification & Validation, Fault Isolation, Technical Documentation

CORE SKILLS – Analytical Problem-Solving, Team Collaboration, Technical Communication, Adaptability

AVAILABILITY

Available full-time from January 2027